Early leafout extends calendar but not thermal growing seasons

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Abstract

Recent results suggest earlier leafout does not always increase plant growth, but have struggled to explain why. Here, using rarely available plant-level phenology from a common garden of 13 woody species, we show that earlier leafout leads to earlier growth cessation, and thus longer but cooler growing seasons, especially for early-leafing species. These results challenge fundamental assumptions about calendar versus thermal growing seasons and growth cessation in woody species, with implications for carbon models

1 Introduction

Terrestrial forests are a major mitigation pathway for climate change, sequestering 20% of greenhouse gas emissions (1). Most models of climate change assume warming will drive longer seasons, more tree growth and, thus, more carbon storage (Figure 4a), following decades of evidence from ecosystem-scale studies (2; 3). However, recent findings from observations of individual trees have questioned this assumption (4; 5). At the same time new experiments on saplings have suggested that plants may adjust their end of season timing dynamically such that longer seasons do not increase total productivity (Figure 4b), though the mechanism—and prevalence—of this effect is unclear (6; 7; 8).

These results suggest fundamental gaps in our understanding of how early-season events and the environment of the growing season shape end-of-season events, but addressing these gaps requires working across plant strategies (e.g., ruderal vs. competitive species) and varying definitions of a growing season. The timing of leafout and senescence are critical components of plant growth strategies they vary strongly across species (9). Within species, populations also vary (10), especially because end-of-season events are usually locally adapted (e.g. budset 11; 12). Increasing evidence also suggests local climate may define narrow periods of growth for many species, with most significant growth earlier in the growing season (8). These results suggest the calendar growing season (i.e., the number of days between the start and end of the growing season) may be less informative to predicting growth than estimates that directly incorporate environmental conditions (13).

The thermal growing season—a period of favorable meteorological conditions for plant growth (13)—offers an alternative to the calendar growing season. It is widely used, most commonly in the metric of growing degree days, as a temperature-derived measure of time that accumulates when temperatures are above a certain minimal threshold (14; 15). Because plant photosynthetic rates increase with temperature (16) the thermal growing season is mechanistically related to primary productivity, and thus may be a better proxy than calendar time for relating growing season length to carbon gain. Yet we generally lack plant-level phenological data for the start and end of season to examine how the calendar and thermal seasons of species and populations compare across years.

Here, we use rarely available plant-level data from three years of a multi-species, multi-population common garden to test how variation in start and end of season events drive shifts in calendar and thermal growing seasons. Our results leverage phenological observations of leafout and growth cessation (budset and leaf coloration) at the individual tree-level to understand how growing seasons scale across populations, species and communities, and may thus enhance our understanding of ecosystem measures to improve forecasting.

Methods

Common Garden

In 2014-2015, we collected seeds from four field sites in northeastern North America spanning approximately a 3.5° latitudinal gradient. The four field sites included Harvard Forest (42.55°N, 72.20°W), the White Mountains (44.11°N, 71.40°W), Second College Grant, (44.79°N, 71.15°W), and St. Hippolyte, QC, CAN (45.98°N, 74.01°W). We transported all seeds back to the Weld Hill Research Building at the Arnold Arboretum in Boston Massachusetts (42.30°N, 71.13°W) where we germinated seeds following standard germination protocols, and grew them to seedling stages in the research greenhouse. In the spring of 2017 we out-planted seedlings to establish the garden.

Plots were regularly weeded and watered throughout the duration of the study and were pruned in the fall of 2020. Survival of *Acer spicatum*, *Acer pensylvanicum*, *Vaccinium myrtilloides*, *Quercus alba*, and *Quercus rubra* was limited and these species are therefore not included in the following analyses. Based on survivorship in the common garden, our subsequent analyses included an average of 14 individuals per species. Our statistical analyses account for the unbalanced design that is frequently occurs in common gardens and other provenance trials.

Phenological monitoring:

For the years of 2018-2019, we made phenological observations of all individuals in the common garden twice per week from February to December. In 2020 due to the COVID 19 pandemic, we monitored once per week from March to November. We describe phenological stages using a modified BBCH scale (17) a common metrics for quantify woody plant phenological progression. We observed all major vegetative stages (budburst BBCH 07, leafout BBCH 15, end of leaf expansion BBCH 19, leaf coloration/drop BBCH 97, reproductive phases flowering BBCH 60-65, fruiting BBCH 72-79 and fruit/cones fully ripe BBCH

89). We added additional phases for budset and labelled full budset as BBCH 102.

Data analysis

To better understand the role that variation in leafout and budset phenology play in determining calendar growing season length among species, populations, and years we fit a Bayesian hierarchical model with a normal (Gaussian) error distribution. We calculated growing season duration by subtracting the day of leafout from the day of budset. We fit an intercept-only model with phenophase timing (leafout, budset, leaf coloration) as the response variable with partial pooling across species, populations, and years. We only included observations with both start and end of season phenophases observed on the same plant for this analysis (n= 595).

To assess the relationship between variation in leafout timing and calendar and thermal growing seasons we fit two additional regression models with thermal or calendar growing season length as the response variable and day of leafout as the main prediction. To account for species-level differences we included partial pooling on the slope and intercept of species.

We define the thermal growing season as the cumulative growing degree day heat sums between the day of leafout and the day of budset for each species. We calculated daily heat sums using the R package "pollen" (18) using a 10 °C base temperature with minimum and maximum daily temperature data from the Arboretum weather data record: https://arboretum.harvard.edu/research/data-resources/weather/. These weather data only ran through August of 2020, so we combined these data with temperate series from the nearest Network for Environment and Weather Applications (NEWA) weather station: "Boston (Weld Hill)" at https://newa.cornell.edu/all-weather-data-query/.

All models were fit in the R package "brms" (19) on 4 chains with a 4000 iteration warm-up and 1000 sampling iterations on each chain for a total of 4,000 sampling iterations across all chains. We evaluated model fit, with no divergent transitions, rhats less than 1.01 and high effective sample sizes. We performed all analyses in R version 4.1.2 (20).

2 Results

We observed high variation in the timing of both start and end of season events with resulting calendar and thermal growing seasons varied almost two-fold across individual trees (2.1x and 1.8x respectively), with earlier leafout leading to longer calendar growing seasons (Figures 4c,d, S3), as routinely seen with anthropogenic climate change (21; 22). We observed a positive correlation between leafout and end of season phases (i.e., budset or leaf coloration) with earlier leafout resulting in earlier budset or leaf coloration (DO WE NEED A FIGURE OR NUMBERS).

Longer calendar growing seasons did not lead to higher thermal growing seasons, as generally assumed; instead early calendar growing seasons led to lower—cooler—thermal growing seasons (Figure 4c,d). These results were similar when we determined the growing season length with leaf coloration instead of budset, Figure S3).

Differences between thermal and calendar growing seasons were strongest in the earlier-leafout species, which leaf out during the coolest part of the spring (Figure 4d)For later leafing species, earlier leafout did not substantially reduce their thermal growing season, bit it also did not increase it—suggesting thermal seasons do not increase with calendar growing seasons for any of our studied species.

Species was the a major predictor of variation in leafout and budset($\sigma_{species}$ days for leafout: 8.1,UI$_{90}$[5.5, 11.6], days for budset: 9.7, UI_{90}[7, 13.6]).Beyond species-level variation, we found high interannual variation in both leafout and budset ((leafout: σ_{year}: 10.7 days, UI_{90}[4.9, 20.9], budset: σ_{year}: 15 days, UI_{90}[6.4, 30.8], Figure 2b,).

We found little variation across populations in leafout ($\sigma_{population}$: 0.6 days, UI_{90}[0, 1.8], Figure 2a).We did find evidence of local adaptation ($\sigma_{population}$: 2.3 days, UI_{90}[0.4, 6.1], Figure 2a), with populations from the site with the fewest frost free days (FFD, Figure S4) setting buds approximately two days before the site with next fewest FFD. These differences, however, were weak (Figure 2a) and 7X and 4X smaller than the variation due to year and species (respectively) We observed similar patterns with leaf coloration as the end of season phase (Figure S5).

3 Discussion

Across three years and 13 co-occurring species (Tab: 1) collected from four populations across a 3.5° latitudinal gradient (Figure S1), phenological observations from our common garden showed that earlier leafout does not extend their thermal growing season—a proxy for potential carbon uptake period—despite extending the calendar growing season. This was driven by a relationship between leafout and budset at the individual level combined with differences in thermal conditions of the early season during leafout and later in the season during budset. When plants leafed out earlier in the spring they stopped growth (budset) earlier, but the additional days of growth in the spring—when temperatures were cool—were rarely offset thermally by the days lost in the later season to earlier budset, which occurred in warmer months (July-August, Figure 3).

These results suggest that some of the apparently conflicting recent findings of whether longer seasons increase growth or productivity may be due to different scales of observation, including the different species they integrate over. Our result that individual trees appear to accelerate their end of season events with earlier start of season events supports other studies conducted on a small number of species (using alternative metrics of senescence, 23; 6). Thus, we may expect that studies of individual trees could find little relationship between earlier leafout and growth (24; 25; 6), which could be driven by currently unstudied shifts in the thermal seasons. Additionally, because differences between thermal and calendar growing seasons were species-dependent, previous findings may depend on the species studied (24; 25). This cross-species variation likely scales up to affect ecosystem-level estimates, which generally have one measure of the start of season—for the earliest leafout species—and one for the end of season, likely capturing the later-leafout species.

The shift in budset with earlier leafout suggests a potential pathway to better understand and predict ecosystem dynamics, given additional research. While most work on budset has stressed the role of photoperiod (12), other studies have found evidence of temperature control (26; 27). Our results suggest an additional potential role of leafout timing, which could be confounded with temperature. Teasing apart these potential cues while also testing these relationships for other end of season events could help resolve connections between start and end of season events with growth. While we found similar findings using leaf coloring, there are many ways that plants transition from growth to dormancy each year (27)—with different implications for carbon gain (13).

The shift in budset with earlier leafout suggests a potential pathway to better understand and predict recent findings, but will require a new approaches to how we view and study budset and other end of season events. Most work on budset in broadleaf species comes from one genus (*Populus spp.*) and has stressed the major control of photoperiod limiting responses to anthropogenic warming. Yet some studies have found evidence of temperature (28; 26; 27), contrasting with this prediction. Teasing apart effects of temperature from intrinsic correlations between start and end of season events appears an important next step. Doing so across additional species may be especially helpful, as a focus on single species may not always predict broad scale trends (29), and our results suggest important variation across species.

Budset represents the stop of primary growth and generally correlates strongly with the stop of radial growth (12), but it is only one of many ways that plants begin to transition from growth to dormancy each year—generally at different times for different events (27). While the need to better understand which metrics of end of season events correlate best with growth and carbon gain has been often discussed (30; 13), our results suggest we also need to study how these events shift with earlier and warmer years. Studies of leaf longevity have begun to examine this, but more work across different metrics of end of season and across many more species is critical.

Given that photosynthesis is limited by cold temperatures in early spring, earlier leafout appears to provide limited opportunity for substantial growth and may especially deprive early-leafing species from fully using late-season warmth. While this may explain why multiple studies have failed to find correlations between longer calendar seasons and increased plant growth (4; 5), it generates a new question of why any species leafs out when thermal conditions are unfavorable. Varying light availability over the early season combined with different plant strategies may alter this equation, however, potentially making this a successful strategy for some species. Supporting this, we found understory species often showed the strongest relationship, suggesting they leaf out in thermally poor conditions to maximize access to light before the upper canopy closes.

The link between earlier leafout and lower thermal seasons was driven by variation across years at the individual and species levels—scales that are not always well studied in phenology research (21). While satellite observations can document intriguing trends (e.g., 8), observations that include far more species at a finer scale are likely critical for mechanistic understanding. Sampling across species of different leafout and growth strategies appears especially important, alongside studies of how different communities and climates may shape the relationship between calendar and thermal growing seasons. Understanding whether thermal responses are consistent across species with different strategies could also inform how the thermal season varies over time and space, improving our ability to accurately model biological carbon fluxes.

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4 Figures

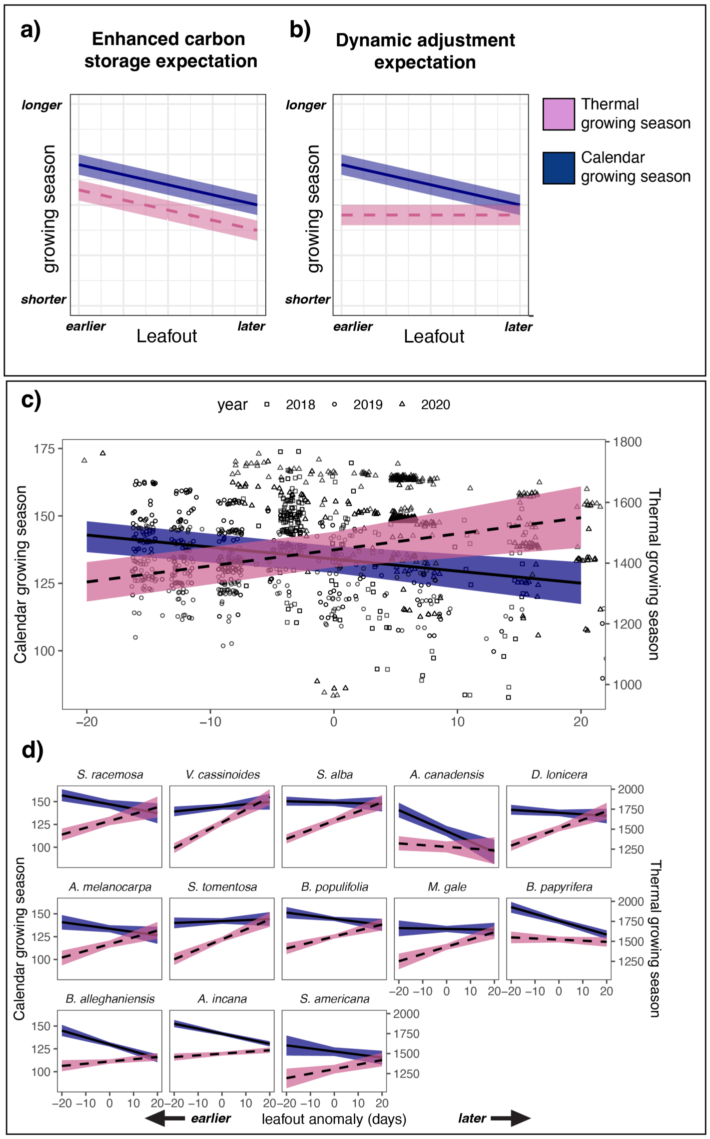

Figure 1: The relationship between leafout timing and growing season length differs between the calendar and the thermal growing seasons. Panels (a) and (b) conceptually demonstrate the expected relationships between leafout timing and both calendar and thermal growing seasons under two different expectations, with (a) showing the expectation that longer seasons lead to increased carbon storage, while panel (b) shows the expectation that plant dynamically adjust the end of their growing season such that longer growing seasons do not increase carbon storage. In our study, later leafout resulted in a shorter calendar growing season (median estimated effect: -0.4 days, $UI_{90}[-0.7, -0.2]$) (c) and this pattern was consistent across species in our study (d). In contrast, later leafout resulted in a longer thermal growing season (median estimates effect: 5.3 growing-degree days, $UI_{90}[2.1, 8.4]$) (d), an effect that was stronger for species that typically leafout earlier in the season—panels in (d) display in the typical order of leafout among species. The black trend lines represent the median effect of leafout timing on growing season length and shaded region 90% uncertainty intervals. Points in (c) represent the raw data (per individual tree per year).

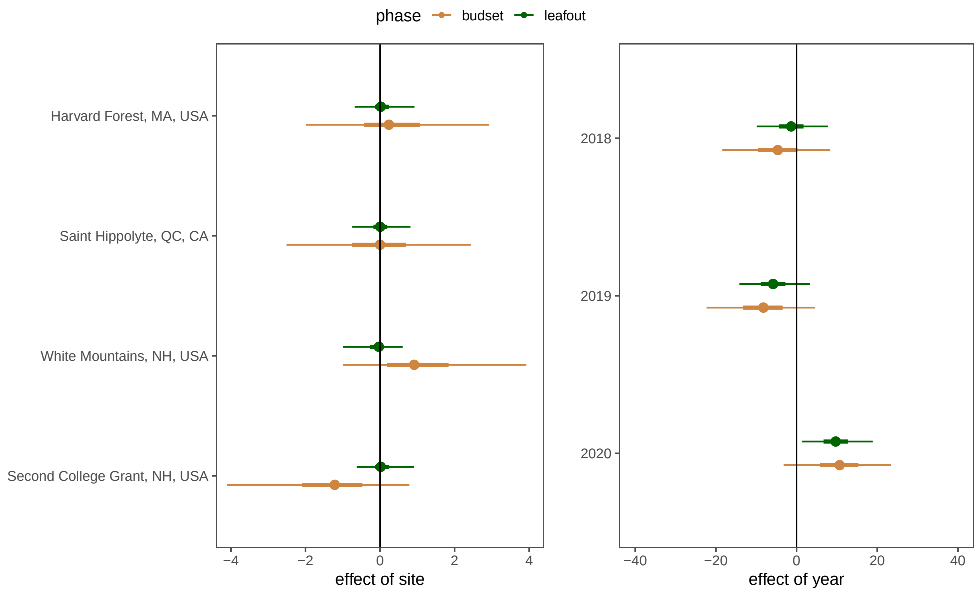

Figure 2: Difference in leafout and butset partitioned between populations (a) and years (b). Points represent the median effect size estimate, and thick and thin bars the 50% and 90% uncertainty intervals.

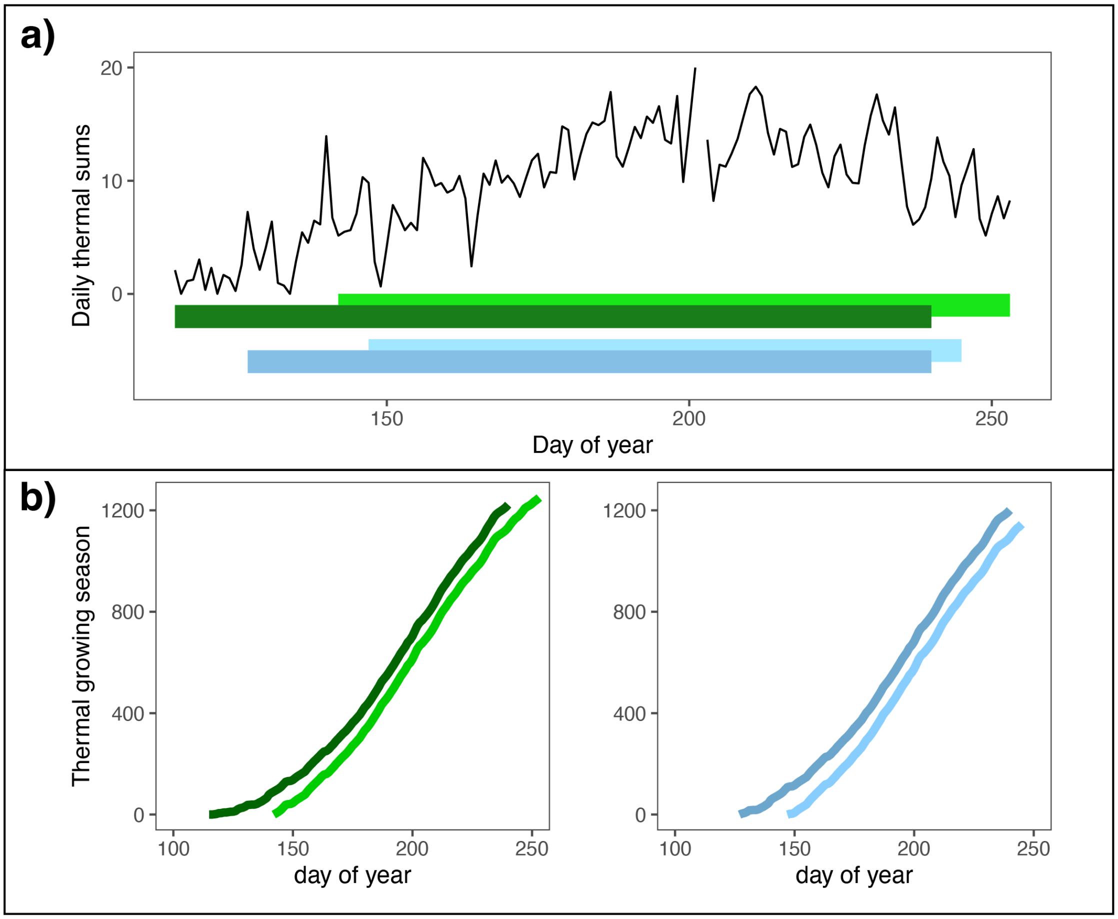

Figure 3: Varying thermal conditions across the calendar growing season combined with correlated leafout and budset can generate complex relationships between the calendar and thermal growing seasons. Panel a) depicts the daily heat sums from the location of a common garden in Boston (MA, USA) in 2019 and the calendar growing season of early and late leafing individuals within two species, one that leafs out earlier, *Aronia melanocarpa* (green bars) and one that leafs out later, *Myrica gale* (blue bars). Panel b) depicts the relationships between the calendar and thermal growing season for each of these individuals. Despite the earlier individual (dark green line) of *Aronia melanocarpa* having a 10 day longer growing season (by leafing out 28 days earlier and setting bud 18 days earlier) than the later leafing individual (light green line), it has a lower (cooler) thermal season because most of its early growth advantage takes place in the thermally unfavorable early spring. In contrast, for *Myrica gale*, both the early (dark blue) and later (light blue) individuals leaf out later in the spring, such that the 11 day longer calendar growing season (from a 15 day leafout and 4 day earlier budset) of the earlier individual results in only slightly warmer thermal season.

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